You said:

What do you say?

RIGHT PANEL:

{
Version nr

Launch Ironwail
}
{
"LEVEL IMPORTER:"
  visual lightmap resolution (in the end just for drawing convenience if
larger than source.)
  target lightmap resolution (ask and inform that higher ones take time to
build while lower ones will decrease quality before setting this) that it will take time and space.
  game /mod
  browse game maps
  import quake.bsp
  import quake .map (possible future feature)
  export bps
}
"SETTINGS:"
installation folder
ericw folder
irowail exec
rerelease exec (possible future feature)
speed
fast mutltiplier
safe settings (possible future feature)
optimize view clipping (is this needed after invoking it on import?)
}
{All the light stuff can for now stay here}
{we probably dont need all the diagnostics anymore. and those that stay are
probably more something for an advaned or debug mode.}

TOOL BUTTONS LEFT:

got to info player start
focus selected face
ilolate selected face
reveal hidden faces
start fps nav
Paint Texture(idea): (enters paint mode. Selected faces are all automatically detached. When leavening paint mode check via hash wich of the
faces actually recieved paint and retach the others. It would be nice to somehow
have a way to higlight painted faces in the level to spot possible ones that got
some unwanted paint.)
Paint Lightmap

Worked for 33s

ChatGPT said:

I like this direction considerably more than the generic editor layout I suggested earlier. It gives Quake Studio a clear purpose: **the level viewport is the centre of the application, the right side handles project configuration, and the left toolbar contains actions that operate directly on the level.**

I'd use your structure as the basis for the workspace, with a few refinements.

1. Right panel — project and configuration

I'd organise it into collapsible sections:

Version and launch

- QSS version
- Launch Ironwail

Level Importer

- Visual lightmap resolution
- Target lightmap resolution
- Game / mod selection
- Browse game maps

- Import `.bsp`
- Export `.bsp`

For target resolution, I agree that we should explain the trade-off before applying a change. In particular, the user should understand that smaller WU/luxel values mean finer lightmaps, longer compile times and larger BSPs. Visual resolution should remain independent: it controls the working presentation, not the source lightmap quality.

I'd keep `.map` import visibly marked as a future feature, rather than presenting a button that cannot yet work.

Settings

- Quake installation folder
- ericw-tools folder
- Ironwail executable
- Speed
- Fast multiplier

I'd leave rerelease support and safe settings for later.

Lighting

- Keep the existing lightmap generation and related controls here for now.

Advanced / Debug

- Move the diagnostics that remain useful here. We can remove redundant diagnostics after reviewing which ones actually provide value.

2. Left toolbar — level operations

Your proposed tools make sense as viewport actions:

- Go to Player Start
- Focus Selected Face
- Isolate Selected Face
- Reveal Hidden Faces
- FPS Navigation
- Paint Texture

- Paint Lightmap

I would make the two paint tools explicit modes, with a clear visual indication when one is active. That prevents accidental painting and makes the interaction model predictable.

Paint Texture: your detach/reattach idea

This is a good approach, but I'd refine the implementation slightly:

1. On entering paint mode, record the selected faces and their original attachment state.
2. Detach the selected faces so texture painting cannot bleed into neighbouring faces.
3. On exit, compare each face's painted image region against its original state.
4. Keep genuinely painted faces detached; restore the original attachment state for untouched faces.
5. Highlight faces that received paint, so accidental changes are easy to spot.

One important detail: a hash can tell us whether image data changed, but not necessarily which face caused the change if multiple faces share an image. We should therefore design the tracking around per-face image regions or isolated paint targets, rather than relying on a single shared-image hash.

3. A few design decisions I'd make

Use Blender's native layout conventions. Your left/right split maps very naturally to Blender:

- **Left toolbar:** viewport tools and modes.
- **Centre:** 3D level viewport.
- **Right sidebar:** QSS project settings and workflow panels.
- **Bottom area (optional):** image/lightmap editor when needed, rather than permanently sacrificing viewport space.

That gives us a genuinely integrated Blender workspace instead of a separate application UI bolted onto it.

Keep the right panel project-oriented. The game/mod and installation settings should persist between sessions. Map-specific choices, such as lightmap resolution, should be associated with the current import/generation operation rather than silently changing global settings.

Keep diagnostics out of the normal workflow. I agree with moving the useful ones into Advanced/Debug. We should retain the underlying validation where it protects imports and exports; hiding diagnostic buttons is not a reason to remove integrity checks.

Two further refinements

- **Optimize View Clipping:** I would make this an automatic import-time operation, not a permanent settings toggle. Add a manual *Recalculate View Clipping* action if it proves useful later.
- **Painted-face highlighting:** Definitely worth doing. A temporary overlay or face-selection colour would make changed faces identifiable without permanently altering their materials.

4. Implementation order

I'd avoid trying to build every feature into the workspace at once.

Stage	Deliverable
1	Quake Studio workspace, left toolbar and right QSS panel
2	Move existing import, export, settings, launch and lighting controls
3	Implement viewport tools and their active states
4	Improve Texture Paint mode with per-face change tracking and highlighting
5	Integrate Lightmap Paint and refine the editor layout
6	Move remaining diagnostics into Advanced/Debug

One additional point: **Paint Texture and Paint Lightmap should be mutually exclusive modes.** Switching between them should cleanly exit the previous mode and restore its temporary state.

My recommendation is to make your proposed layout the actual product design, not just another arrangement of existing panels. We can start with the workspace shell and migrate the existing functionality into it, preserving the tested import/export and lighting code.

And yes: the old sidebar should remain available as a temporary fallback until the new workspace has passed real-map testing.

You said:

Sound all very good. Setting you to think! Lets see how many Stages you accieve. :D Impress me!

Worked for 11m 22s >

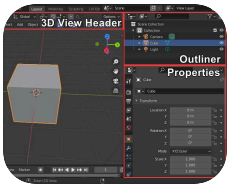
ChatGPT said:

CI passed 149 tests passed

First Quake Studio workspace implementation is committed. 😁

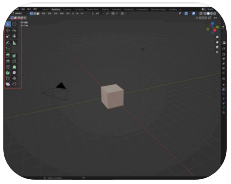
I got the workspace shell, the main workflow migration and the first set of viewport tools into this release.

What's in v0.17.0



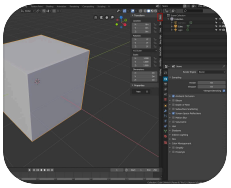
1. Dedicated Quake Studio workspace

Creates a separate workspace by duplicating the current one. Your original Layout and its editor arrangement remain intact.



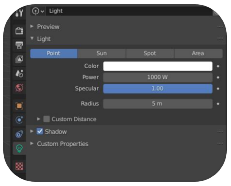
2. Left toolbar

Player Start, focus selection, FPS navigation, isolate/reveal faces, Paint Texture and Paint Lightmap.



3. Right-side workflow panel

Import, game/mod selection, map browsing, export, Ironwail, installation settings, navigation settings and lighting controls.



4. Lighting and advanced controls

Lighting Controls is its own collapsible section. Diagnostics, audits and recovery tools are under Advanced / Debug.

Important improvements

- Visual lightmap resolution and target WU/luxel density are now separate settings.
- The target density displays a quality, compile-time and file-size explanation.
- Map browsing uses the selected visual resolution.
- View clipping remains automatic on import.
- Paint Texture now detaches selected faces before entering Texture Paint.

The automatic detection of actually painted faces, reattachment of untouched faces and painted-face highlighting are **not implemented yet**. Those deserve their own stage rather than a rushed approximation.

Get v0.17.0

Quake Surface Studio 0.17.0 Ready for Blender testing

- Python compilation: passed
- Standalone tests: 149 passed
- Blender install package: assembled successfully

Download Blender install artifact ↗

CI run #219 · commit `d4c3fea`

What I want to test next

1. Install v0.17.0 alongside your current setup.
2. Open the old Quake sidebar and click **Open Quake Studio Workspace**.
3. Check that the new workspace opens, the left toolbar tools appear and the Quake Studio tab is visible on the right.
4. Import e3m4 and test the viewport actions, painting, lighting and export.

The CI cannot run Blender's actual graphical workspace interactions, so I want your Blender 5.2 feedback before calling the workspace migration complete.

The feature branch is still isolated: `main` remains untouched, and PR #1 remains unmerged.

I'm looking forward to seeing this running in your Blender setup. This is the point where QSS starts becoming an actual Quake editing environment rather than just an add-on panel. 😁

You said:

3. workspace opens but i dont see a left toolbar. Do i need to be in object or edit mode?

ChatGPT said:

You can use both modes; the available QSS tools change depending on the mode.

- **Object Mode:** Go to Player Start, Focus Selected Face and FPS Navigation.
- **Edit Mode:** Focus Selected Face, Isolate Selected Faces, Reveal Hidden Faces, Paint Texture and Paint Lightmap.

First, move your mouse over the 3D viewport and press `T`. That's Blender's shortcut for toggling the left toolbar. (`N` toggles the right sidebar.)

If the toolbar appears but contains no QSS tools, or remains completely absent, tell me. That would indicate a workspace/tool-registration issue in v0.17.0, which I'll fix rather than make you fight Blender's UI. 😁

You said:

Ah! The toolbar appears but with none of the addonspecific tools.

Worked for 2m 55s

ChatGPT said:

Found and corrected the likely cause! 😁

In v0.17.0, I had accidentally put all QSS tools into nested toolbar groups. The first icon also looked almost identical to Blender's regular selection tool. Not exactly the toolbar experience we wanted.

v0.17.1 fixes this:

- Each QSS action now gets its own visible toolbar button.
- Object Mode and Edit Mode tools are registered separately.
- Both groups are placed directly after Blender's selection tools.
- Tool invocation now follows Blender's standard WorkSpaceTool keymap format.

CI passed **149 tests passed**

Quake Surface Studio v0.17.1

Download Blender installer ↗

Run #224 · commit `427caded`

Please install this version, disable/re-enable QSS, and reopen Quake Studio. The QSS buttons should now be individually visible near the top of the left toolbar.

I haven't been able to verify the graphical toolbar in Blender itself, so your next test will confirm whether this fixes it.

You said:

Still no qss buttons

Worked for 2m 55s

ChatGPT said:

Okay, this time I changed the registration strategy rather than just rearranging the buttons. 😁

I checked Blender 5.2's actual toolbar implementation. Its 3D View toolbar combines a shared tool list with mode-specific lists. QSS now registers its tools in that shared list, making them available across Object, Edit and Paint modes. It also removes stale registrations left by the earlier versions. Blender Proj… +1

CI passed **149 tests passed**

v0.17.2 — toolbar migration fix

Download the updated Blender installer ↗

There is also a new diagnostic in the Quake Studio panel:

Viewport tools registered: 8/8

Expected display after successful registration

Refresh New control

Please install v0.17.2, disable and re-enable QSS, and reopen Quake Studio.

The registration counter is important:

- **8/8:** Blender accepted all eight QSS tool definitions.
- **Anything less:** We have a registration problem, and the counter should help identify it.

I haven't tested the graphical result in Blender 5.2 myself, so I won't claim this is fixed until you confirm the buttons actually appear. `main` remains untouched and the PR remains unmerged.

You said:

Pasted text(2026...
Document

when pressing open workspace buttonsa appear left and then:

Worked for 3m 4s

ChatGPT said:

Got it. Your log finally exposes the actual bug. And this one is mine.

The error is not a missing toolbar, a Blender mode issue or an installation problem. QSS has inserted a nested tuple into Blender's internal toolbar definition list. Blender expects a tool definition there, but receives a tuple instead:

```
AttributeError: 'tuple' object has no attribute 'idname'
```

It happens repeatedly while Blender draws the toolbar and tries to activate tools. The log also confirms that the workspace itself was created successfully. Pasted text(20261109-041824).txt

Pasted text(20261109-041824).txt

So v0.17.2 made the toolbar worse. I'm going to repair the list structure itself, remove malformed entries left by previous QSS versions, and stop using Blender's `after` insertion mechanism where it can create nested groups.

I found the culprit and fixed the actual data corruption. This time the traceback gave us enough evidence to identify exactly what Blender was rejecting.

What was wrong

Blender's toolbar supports individual tool definitions and one-level tool groups. Our earlier registration code accidentally created nested groups. The old cleanup routine also preserved those malformed groups instead of removing them.

That is why every attempt to draw the toolbar produced the same exception.

v0.17.3 — structural repair

CI passed 149 tests passed

The new version:

1. Removes stale QSS entries from all toolbar mode lists.
2. Flattens malformed nested groups left by previous versions.
3. Registers each QSS tool as an individual Blender tool definition.
4. Avoids the problematic `after` insertion mechanism entirely.
5. Counts registered tools across all relevant modes.

Install the corrected version

Quake Surface Studio v0.17.3

149 / 149

Standalone tests passed

Download Blender installer ↗

CI run #238 · commit `a79bfe6`

My recommendation for installation: install v0.17.3, then restart Blender. This gives us a clean toolbar state before QSS registers its repaired tools.

After opening Quake Studio, look at the right panel. It should report:

```
Viewport tools registered: 8/8
```

Then check the left toolbar in Object Mode and Edit Mode.

I have also updated PR #1 ↗ to reflect the workspace work. It remains a draft, unmerged, and `main` is untouched.

One honest caveat: CI validates the code and all 149 tests, but the final graphical confirmation still needs your Blender 5.2 session.

We finally have a specific fix for the exact exception in your log, rather than another toolbar guess. 😁

You said:

```
bpy.ops.object.delete(use_global=False, confirm=False)
Deleted 1 object(s)
'str' object has no attribute 'idname'
Python: Traceback (most recent call last):
  File "/app/blender/5.2/scripts/startup/bl_ui/space_view3d.py",
line 65, in draw
    self.draw_tool_settings(context)
    ~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^
  File "/app/blender/5.2/scripts/startup/bl_ui/space_view3d.py",
line 78, in draw_tool_settings
    tool = ToolSelectPanelHelper.draw_active_tool_header(
        context, layout,
        tool_key=('VIEW_3D', tool_mode),
    )
  File
"/app/blender/5.2/scripts/startup/bl_ui/space_toolsystem_commo
n.py", line 813, in draw_active_tool_header
    item, tool, icon_value = cls._tool_get_active(context, space_type,
mode, with_icon=True)

~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
^^^^^^^^^
  File
"/app/blender/5.2/scripts/startup/bl_ui/space_toolsystem_commo
n.py", line 326, in _tool_get_active
    if item.idname == tool_active_id:
       ^^^^^^^^^^^
AttributeError: 'str' object has no attribute 'idname'
Python: Traceback (most recent call last):
  File
"/app/blender/5.2/scripts/startup/bl_ui/space_toolsystem_commo
n.py", line 752, in draw
    self.draw_cls(self.layout, context)
```

```
    ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^
  File
"/app/blender/5.2/scripts/startup/bl_ui/space_toolsystem_commo
n.py", line 710, in draw_cls
    is_active = (sub_item.idname == tool_active_id)
                 ^^^^^^^^^^^^^^^
AttributeError: 'str' object has no attribute 'idname'
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```

```
mode, with_icon=True)

~~~~~~~~~~~~~~~~~~~~^^^
```

```
"/app/blender/5.2/scripts/startup/bl_ui/space_toolsystem_common.py", line 326, in _tool_get_active
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       ^^^^^^^^^^^
AttributeError: 'str' object has no attribute 'idname'
Python: Traceback (most recent call last):
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  File
"/app/blender/5.2/scripts/startup/bl_ui/space_toolsystem_commo
n.py", line 710, in draw_cls
    is_active = (sub_item.idname == tool_active_id)
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Worked for 3m 7s

ChatGPT said:

I found the gap in v0.17.3: its cleanup removes QSS tools and flattens nested tuples, but it treats **any other value** as a valid toolbar entry. A stray string therefore survives cleanup and Blender later tries to read its `.idname`.

The repair needs to be stricter: keep only actual ToolDef-like objects, preserve legitimate separators, and discard unsupported values such as strings from every mode list. This is still a toolbar data-structure issue; the object deletion itself succeeded.

The object deletion succeeded. The error occurs immediately afterward when Blender redraws the viewport toolbar and encounters an invalid string entry.

v0.17.4 CI passed

The fix is now committed to the feature branch.

The new cleanup:

- Removes strings and other values that are not valid ToolDefs.
- Recursively repairs nested tuple and list groups.
- Removes stale QSS tools and normalizes separators.
- Uses the same strict validation when counting registered tools.

Validation: Python compilation passed and all 153 tests passed, including four new toolbar regression tests.

Install v0.17.4: Download Blender install artifact ↗

SHA-256: `a30f67b44e6e6e45d3ae96db0d0a4c0405e46c81622b85a6b8d05e744845958f`

The PR remains open and in draft ↗. `main` is unchanged.

Please fully restart Blender after installing v0.17.4. The Quake Studio panel should report **8/8 viewport tools registered**. I haven't yet verified the toolbar in the actual Blender GUI, so that remains the final check.

You said:

> toolbars are there now. changes between object and edit. The standard toolbars are removed i think wich is bad especially in paint mode. Not all buttons work... Take some time, think it all through and make it work.

Worked for 9m 27s

ChatGPT said:

I found the root cause and corrected the toolbar integration in **QSS v0.17.5**.

The main mistake was in my previous fix: Blender 5.2's `ToolDef` is a named tuple, so the cleanup code was accidentally dismantling Blender's own tool definitions. It also removed callable entries that Blender uses to populate mode-dependent tools, including Paint-mode selection tools. That was why native buttons disappeared and why the earlier string errors occurred.

What changed

Native Blender toolbars

- Preserves the built-in Object, Edit, Texture Paint, Vertex Paint, Weight Paint and Sculpt tools.
- Preserves Blender's dynamic tool generators and existing tool groups.
- Removes only stale QSS entries and genuinely malformed data.
- Uses Blender's registration and unregistration lifecycle, including proper cleanup when Refresh is used.

QSS actions

- Object Mode: Player Start, Focus and FPS Navigation.
- Edit Mode: Focus, Isolate, Reveal, Paint Texture and Paint Lightmap.
- Isolate and Reveal now explicitly use a valid 3D View context.
- FPS Navigation is explicitly invoked as a modal operation.

One important usability distinction: the left toolbar selects tools. Selecting a QSS tool and then clicking in the viewport runs it. I added a **Quick Actions** section to the Quake Studio sidebar for direct, one-click commands. When entering Texture Paint, Blender's normal paint toolbar should remain available.

Validation

CI passed **155 tests passed**, including six toolbar-specific regression tests. Python compilation and package assembly also passed.

QSS v0.17.5 — Blender install

Download the Blender-ready add-on ↗

SHA-256

`a79c483ad50c1693a4aece7b9bc52c20e9d0dbe4d3ff15bf8bdf8fd34fd991d1`

Important installation step

Please **close Blender completely** before installing v0.17.5. The v0.17.4 cleanup already damaged Blender's built-in toolbar definitions in the running process. Reloading the add-on alone cannot restore those definitions; a full restart reloads Blender's original tool lists.

The PR is updated ↗, still draft and unmerged. `main` remains untouched.

I have not claimed graphical verification: the remaining check is your Blender 5.2 test. In particular, check that the standard Paint toolbar is back, then try the sidebar Quick Actions for Isolate, Reveal, Paint Texture and Paint Lightmap.

You said:

focus selected face does not work

isolate selected faces works in the way that you klick the button and then
somewhere in the viewport and the selected faces are getting isolated. weird
with the viewport klick. Also think it should toggle the isolation. I guess
that would make reveal hidden faces obsolete?

paint texture and paint lightmap work very good. Here it feels ok that i have
to klick the viewport first but i still think its not necessary.a

Worked for 11m 37s

ChatGPT said:

Implemented the interaction changes in **Quake Surface Studio v0.17.6**.

164 tests passed

Focus Selected Faces

Reworked Focus to calculate the world-space bounds of the selected BSP faces and frame them directly in the active 3D View.

- In Edit Mode, it frames selected faces (or selected vertices if no faces are selected).
- In Object Mode, it frames the complete BSP object.
- It no longer relies on Blender's context-sensitive `view3d.view_selected` operator.

Toggle Face Isolation

Isolation now works as a proper toggle:

1. Select the faces to keep visible.
2. Click **Isolate Selected Faces** in the Quake Studio sidebar.
3. Click **Restore Visibility** to restore the previous hidden faces and selection.

The previous state is saved on the BSP object, so it survives saving and reopening the `.blend` file. It uses BMesh's `hide_set()` method to update the associated geometry visibility correctly. Blender Python...

I removed the separate QSS Reveal button. Blender's native **Alt+H** remains available for revealing geometry hidden by other operations.

Toolbar interaction

The left toolbar now contains:

- FPS Navigation in Object Mode.
- Paint Texture and Paint Lightmap in Edit Mode.

Player Start, Focus and Toggle Isolation are direct sidebar actions. They no longer require clicking an arbitrary point in the viewport after selecting a toolbar tool. The paint tools also remain available as direct sidebar actions.

Installation

QSS v0.17.6 — Blender install

Download the Blender-ready add-on ↗

SHA-256

`a7a6f539ee7dedaee1b614c8aa0448c94d696811e7ed643d447d3559abe1c477`

CI passed Python compilation and all **164 tests**. The new focus and BMesh visibility behavior still need confirmation in the actual Blender 5.2 interface.

The PR is updated ↗, still draft and unmerged. `main` is unchanged.

Please test Focus on a single face, then isolate and restore a selection that includes some previously hidden faces. That second test is particularly useful for checking that the toggle restores the original state rather than simply revealing everything.

You said:

focus and isolate work. And the fps mode is still very clumsy. And i cant deactivate it after activating. Do some research. There are working implementations for blender im shure. I like the viewport controls from trenchbroom for instance. Its kind of what i wanted only they press lmb instead of mmb. Or the fly movement from quakes nocip. Something like that.

Worked for 13m 16s

ChatGPT said:

Blender 5.2 already has a native mode that matches the core of what you describe: **Walk Navigation**. It gives you continuous mouse-look, `W/A/S/D` movement, `Q/E` vertical movement, and temporary speed boost with Shift. Crucially, **LMB confirms and exits; Esc or**

RMB cancels and restores the starting view. That is much closer to Quake noclip than QSS's current split system, where the mouse only orbits and the keyboard moves.

Blender 5.2 ... +1

I'm going to base the replacement on Blender's own navigation operator rather than maintain a second, competing modal camera controller.Implemented **QSS v0.17.7** on the existing feature branch.

New Quake Walk / Noclip controls

The sidebar's **Start Quake Walk (Noclip)** button now invokes Blender 5.2's native Walk Navigation instead of QSS's custom movement controller.

Input	Action
Mouse movement	Look around
W / S	Forward / backward
A / D	Strafe
Q / E	Down / up
Shift	Move faster
LMB	Confirm and exit
Esc or RMB	Cancel and restore the starting view

It runs without gravity or BSP collision, so it behaves like free-flight noclip. QSS converts its Quake-units-per-second speed setting into Blender's native Walk speed using the imported map's unit scale. Blender 5.2 ... +1

The important deactivation fix: FPS navigation is no longer a persistent left-toolbar tool. It starts from the sidebar and exits through Blender's normal Walk controls. Once exited, the previously selected viewport tool works normally again. No MMB capture or custom timer-driven navigation loop remains.

TrenchBroom's default 3D controls are split between RMB mouse-look, MMB camera translation, scroll-wheel movement and keyboard movement. The new QSS mode instead follows Blender's native first-person Walk model, which is closer to the Quake noclip behaviour you mentioned. trenchbroo... +1

Build and status

CI passed **161 tests passed**, including compilation and package assembly. The new Blender interaction still needs your graphical test; CI does not run Blender's GUI.

Download QSS v0.17.7 ↗

SHA-256: `5f465658ac01426ca359548841a9c8a7f5d41383930dadfc0f98411d32dde68d`

Draft PR #1 ↗ is updated. It remains unmerged, and `main` is unchanged.

The key thing to test is whether mouse-look and WASD now feel natural, and whether a single LMB cleanly returns you to normal selection.